NILESH VISHWAKARMA

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Technical Skills

Programming Languages: C++, JavaScript , SQL
Frameworks & Libraries: React.js, Node.js, Express.js , PostgreSQL , RESTful APIs, Tailwind CSS
Developer Tools: Git, GitHub, VS Code, Docker
AI Tools & Workflow: Cursor AI, Github Copilot, ChatGPT, Grmini, v0
Core Computer Science: Data Structures & Algorithms(DSA), OOP, DBMS, Operating Systems, Computer Networks

Experience

Google Summer of Code (GSoC) Contributor **Mar. 2026 – Present**
Neuroscience Information Unit (NIU) – BrainGlobe Ecosystem *Remote*

- Selected for GSoC 2026 under the BrainGlobe ecosystem to expand trajectory characterization in the `movement.kinematics` module[cite: 3, 5, 20].
- Engineered high-performance, vectorized trajectory analysis algorithms (Straightness Index, Tortuosity, and Sinuosity) using **NumPy** to process massive behavioral tracking datasets efficiently[cite: 40].
- Architected seamless integration with **xarray.Dataset** accessors, enabling researchers to compute kinematic metrics directly while preserving spatial-temporal metadata.
- Implemented a rigorous test suite using **Pytest** for edge-case validation and cross-verified calculations against R's *trajr* package to ensure 100% numerical consistency.
- Configured automated multi-platform continuous integration pipelines via **GitHub Actions** across Linux, macOS, and Windows.

Open Robotics & JdeRobotRemote

- Submitted technical proposals for GSoC 2026 focused on C++ robotics middleware and high-performance Hardware Abstraction Layer (HAL).
- Designed scalable architectures and system designs for real-time distributed robotics applications.
- Optimized low-level control interfaces and modular components to ensure high performance and reliability.

Projects

Zero Waste Food | *React.js, Node.js, JavaScript, PostgreSQL* | Live Demo **June 2026**
∗ Developed a full-stack food donation platform connecting Restaurants, Donors, NGOs, and Admins to reduce food waste and improve food distribution.
∗ Implemented secure role-based authentication, donation management, food claiming, image uploads, expiry tracking, and real-time availability updates.
∗ Designed responsive dashboards using React.js and integrated REST APIs with PostgreSQL to manage the complete donation workflow.
∗ Built an intuitive user interface focused on accessibility, responsiveness, and efficient food distribution across multiple user roles.

PTSRIET-PULSE & Academic Portal | *React, Node,Express, PostgreSQL,Tailwind* | Live Demo **2026**
∗ **Centralized Digital Portal:** Developed a comprehensive academic platform enabling students to access semester-wise curricula, syllabi, and master cheatsheets.
∗ **Admin Control Center:** Engineered an admin dashboard to dynamically manage course materials, enquiries, events, job postings, and curriculum access locks.

Achievements & Certifications

Open Source Contributor: Successfully contributed to Open Robotics,Jderobot, and Neuroinformatics Unit through merged GitHub Pull Requests.
UHack 4.0 National Hackathon: Received Certificate of Merit for the STDAMS project.

Education

Pt. Sukhraj Raghunathi Institute of Education & Technology **Sep. 2024 – Present**
Bachelor of Computer Applications (BCA) (6.708CGPA) *Pratapgarh, Uttar Pradesh*